Curriculum vitae

Mohamed Shamseldein, Ph.D., P.Eng., Senior Member, IEEE
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Research and professional profile

Power-system researcher and engineering leader combining more than thirteen years at IESO with academic research at Ain Shams University. Research connects Grid for AI, AI for the Grid, hydrogen and defence-related resilience, with developing work on post-quantum cyber-physical security. Shared foundations include physical modelling, verifiable decisions and reliable connection requirements.

Selected research contributions

AI for the Grid. Published work on Decision Transformer control, safety-aware dynamic-security assessment, physics-consistent power flow and uncertainty-aware feasibility screening [J5, J6, J8, J9].

Grid for AI. Published studies of data-centre ride-through, storage coordination, fast frequency response and oscillation damping; recipient of the 2026 Energy Storage Canada research award [J3, J4, J7].

Hydrogen integration. Multi-timescale assessment and performance requirements for hydrogen hubs and co-located computing, storage and hydrogen interfaces [J1, J2, J10].

Defence-related resilience. Two accepted papers presented at ITC-Egypt 2026 on grid-infrastructure resilience and stability-constrained integration of pulsed loads [C2, C3].

Quantum and cyber-physical security. Developing work on post-quantum trust dependencies in energy-system control; revised journal manuscript submitted. Focus: cryptographic trust and control integrity.

Awards and recognition

Foundational Research in Energy Storage Award, Energy Storage Canada, 2026. Recognized for research on coordinated battery-storage control supporting data-centre grid reliability and continuity of critical computing loads. Award announcement

Excellence Award, Energy Systems Integration Group (ESIG), 2026. One of five named engineers recognized for collaborative large-load integration and technical knowledge sharing. Award announcement

IESO Beacon Award for Outstanding Leadership, Charge Award for Exceptional Performance, and Spark Award for Exceeding Project Expectations. MITACS Accelerate Ph.D. Fellowship with Hydro One Networks, 2010 to 2012.

Education and credentials

Ph.D., Electrical and Computer Engineering, University of Waterloo, Canada, 2012.
M.Sc., Electrical Power Engineering, Ain Shams University, Egypt, 2007.
B.Sc., Electrical Power Engineering, Ain Shams University, Egypt, 2003.
Ph.D. thesis: Novel Designs for Photovoltaic Arrays to Reduce Partial Shading Losses and to Ease Series Arc Fault Detection.

Professional Engineer, Ontario; Senior Member, IEEE; IEEE Power & Energy Society member. Fundamentals of University Teaching Certificate, University of Waterloo, 2012. Deep Learning Specialization, DeepLearning.AI, 2025.

Appointments and experience

Principal Power System Engineer, IESO, Canada | 2026 to present
Lead technical work on large computational loads, including performance and modelling requirements; coordinate connection-assessment perspectives and mentor engineers.

Power System Engineering Roles, IESO, Canada | August 2013 to 2026
Progressive responsibilities in connection assessments, technical reviews and operating limits for generation, battery storage, transmission and large loads. Conducted and guided reliability studies using PSS/E, PSCAD and PowerTech DSA.

Assistant Professor, Ain Shams University, Egypt | 2016 to present
Department of Electrical Power and Machines, Faculty of Engineering. On approved leave while based in Canada and conducting academic research; IESO is the primary employer. Current academic rank: Assistant Professor; promotion review is in progress.

Lecturer and Research and Teaching Assistant, University of Waterloo | 2008 to 2013
Taught Power Electronics and Motor Drives Analysis; supported engineering courses and laboratories, advised a capstone project and conducted photovoltaic research.

Research and Teaching Assistant, Ain Shams University | 2003 to 2008
Supported electrical-engineering tutorials and laboratories; researched wind-generator fault response.

Professional service and industry guidance

Lead author, Reliability Impacts of Large Loads, ESIG, 2026. NERC drafting-team member for Project 2023-05, Modifications to FAC-001 and FAC-002, representing IESO. Acknowledged contributor to NERC's March 2026 white paper on emerging large loads and reliability standards.

Reviewer for IEEE Transactions on Power Systems, IEEE Transactions on Smart Grid and IEEE Transactions on Sustainable Energy.

Teaching and mentoring

University instruction. Delivered Power Electronics and Motor Drives Analysis (MTE 420) at Waterloo, including lectures, assignments, assessment and student support. Earlier instructional work covered tutorials and laboratories in circuits, electrical machines, power electronics, power systems and protection.

Student guidance. Advised a four-student Waterloo capstone project in 2012 and provided technical guidance to graduate researchers. Professional mentoring at IESO covers large-load connection assessment, facility behaviour, modelling assumptions, study interpretation and the evidence supporting reliability decisions, with related work on storage and transmission.

IESO course development. Developed Integrating Data Centres into Ontario's Power System, with classroom and self-paced materials for junior through principal engineers. Resources include narrated lessons, interactive diagrams, worked examples and knowledge checks. Prepared for internal review and pilot delivery.

Independent education. Author and instructor of the published Udemy course Data Center Grid Integration for Power Engineers. Produce public educational videos on power-system behaviour and emerging-load integration.
Udemy course | YouTube channel

Published articles

[J1] M. Shamseldein, "Grid Integration of Gigawatt-Scale Hydrogen Hubs: A Multi-Timescale Stability Analysis and Connection Requirements for Weak Grid Environments," CIGRE Science & Engineering (CSE), no. 42, October 2026. Published full text

[J2] M. Shamseldein, "Co-Located AI data centers, hydrogen production, and battery storage: a multi-timescale grid-connection assessment," Global Energy Interconnection, advance online publication, 2026. https://doi.org/10.1016/j.gloei.2026.08.002

[J3] M. Shamseldein, "Networked Cyber-Physical Control of Data Centres for Fast Frequency Response via DVFS and UPS-BESS Coordination," IET Smart Grid, vol. 9, no. 1, e70117, 2026. https://doi.org/10.1049/stg2.70117

[J4] M. Shamseldein, "Hyperscale Data Centers as Wide-Area Damping Actuators for Low-Inertia Power Systems," Smart Grids and Sustainable Energy, vol. 11, article 43, 2026. https://doi.org/10.1007/s40866-026-00358-5

[J5] M. Shamseldein, "Generative Autonomous Grid Control: Integrating Decision Transformers with a Two-Stage Safety Stack," IEEE Transactions on Smart Grid, early access, 2026. https://doi.org/10.1109/TSG.2026.3714568

[J6] M. Shamseldein, "DSA-SafetyBench: A safety-aware benchmark for artificial-intelligence-based dynamic security assessment in energy systems," Energy and AI, vol. 25, article 100811, 2026. https://doi.org/10.1016/j.egyai.2026.100811

[J7] M. Shamseldein, "Mitigating Grid Stability Risks From Large AI Data Centres: A Three-Mode Supervisory Control Framework for Centralized UPS Systems," IET Generation, Transmission & Distribution, vol. 20, e70369, 2026. https://doi.org/10.1049/gtd2.70369

Published articles continued

[J8] M. Shamseldein, "Learned Surrogates for AC Feasibility in Stochastic Unit Commitment: A Comparative Study With Conformal Guarantees," IET Generation, Transmission & Distribution, vol. 20, e70345, 2026. https://doi.org/10.1049/gtd2.70345

[J9] M. Shamseldein, "A hybrid GNN-LSE method for fast, robust, and physically-consistent AC power flow," Electric Power Systems Research, vol. 252, article 112380, 2026; first online October 31, 2025. https://doi.org/10.1016/j.epsr.2025.112380

[J10] M. Shamseldein, "Impedance-passivity coordination of hydrogen-coupled AI data centers for oscillation damping and fault ride-through at transmission interfaces," Electric Power Systems Research, vol. 262, article 113557, 2027; first online June 29, 2026. https://doi.org/10.1016/j.epsr.2026.113557

[J11] M. Shamseldein, "Scarcity-era large-load interconnection governance: Reliability enablement, lifecycle tail risk, and benchmarking safeguards," Energy Policy, vol. 215, article 115347, 2026. https://doi.org/10.1016/j.enpol.2026.115347

[J12] M. Shamseldein, "Deep reinforcement learning-based load-side resonance forcing via hyper-scale data center load modulation," IET Smart Grid, vol. 9, e70083, 2026. https://doi.org/10.1049/stg2.70083

[J13] M. Shamseldein, "The split-station architecture: Decoupling transformation from distribution via medium-voltage superconducting virtual bus extensions," IET Electric Power Applications, vol. 20, e70171, 2026. https://doi.org/10.1049/elp2.70171

[J14] M. Shamseldein, "Unconstrained model and merge sorted serpentine arrangement method for dynamic reconfiguration of large PV arrays," Energy Conversion and Management: X, vol. 16, article 100320, 2022. https://doi.org/10.1016/j.ecmx.2022.100320

Published articles continued

[J15] M. Shamseldein, "A fast holomorphic embedding power-flow approach for meshed distribution networks," International Transactions on Electrical Energy Systems, article 9561385, 2022.
https://doi.org/10.1155/2022/9561385

[J16] M. Z. Shams El-Dein, M. Kazerani and M. M. A. Salama, "Optimal photovoltaic array reconfiguration to reduce partial shading losses," IEEE Transactions on Sustainable Energy, vol. 4, no. 1, pp. 145-153, 2013.
https://doi.org/10.1109/TSTE.2012.2208128

[J17] M. Z. Shams El-Dein, M. Kazerani and M. M. A. Salama, "An optimal total cross-tied interconnection for reducing mismatch losses in photovoltaic arrays," IEEE Transactions on Sustainable Energy, vol. 4, no. 1, pp. 99-107, 2013.
https://doi.org/10.1109/TSTE.2012.2202325

[J18] A. A. El-Sattar, N. H. Saad and M. Z. Shams El-Dein, "Dynamic response of doubly fed induction generator variable-speed wind turbine under fault," Electric Power Systems Research, vol. 78, no. 7, pp. 1240-1246, 2008.
https://doi.org/10.1016/j.epsr.2007.10.005

[J19] A. A. El-Sattar, N. H. Saad and M. Z. Shams El-Dein, "Fuzzy Logic Controller for Doubly Fed Induction Generator Driven By Variable Speed Wind Turbine," Scientific Bulletin, Faculty of Engineering, Ain Shams University, Part II, vol. 42, no. 1, pp. 671-683, 2007.

Accepted manuscripts

[A1] M. Shamseldein, "Solver-Consistent Neural Surrogate for EMT-Grade LVRT Margin Screening of Inverter-Dominated Large-Load Interconnections," Journal of Modern Power Systems and Clean Energy. Accepted for publication September 9, 2026.

[A2] M. Shamseldein, "Adversarial Self-Improvement for Resilient Smart Grid Dispatch Control Under Extreme N-2 Contingencies," IET Smart Grid. Accepted for publication, 2026.

Conference papers

[C1] M. Shamseldein, F. Gao and M. Bayoumi, "Integration of Large Loads to the Ontario Power System," CIGRE Energy Forum 2026, Calgary, Canada, September 21-24, 2026, paper 10502, pp. 1-10. Presented September 23, 2026.

[C2] M. Shamseldein, "Stability-Informed Defense Prioritization for Power Grid Protection Against Coordinated Drone Attacks," International Telecommunications Conference (ITC-Egypt 2026), Cairo, Egypt, July 27-30, 2026, paper 1571280128. Accepted and presented by the author.

[C3] M. Shamseldein, "Stability-Constrained Siting of Pulsed Counter-UAS Loads in Transmission Substations," International Telecommunications Conference (ITC-Egypt 2026), Cairo, Egypt, July 27-30, 2026, paper 1571280597. Accepted and presented by the author.

[C4] A. Mahmoud, M. Shamseldein, H. Hasanien and A. Abdelaziz, "Photovoltaic Array Reconfiguration to Reduce Partial Shading Losses using Water Cycle Algorithm," 2019 IEEE Electrical Power and Energy Conference (EPEC), Montreal, Canada, pp. 1-6, 2019.
https://doi.org/10.1109/EPEC47565.2019.9074818

[C5] M. Z. Shamseldein, A. Y. Abdelaziz and A. A. Mahmoud, "Fault Analysis and Detection in Total Cross Tied Photovoltaic Arrays," 2018 Twentieth International Middle East Power Systems Conference (MEPCON), pp. 185-189, 2018.
https://doi.org/10.1109/MEPCON.2018.8635289

[C6] M. Z. Shams El-Dein, M. Kazerani and M. M. A. Salama, "Optimal total cross tied interconnection for photovoltaic arrays to reduce partial shading losses," 2012 IEEE Power and Energy Society General Meeting, pp. 1-6, 2012.
https://doi.org/10.1109/PESGM.2012.6344845

[C7] M. Z. Shams El-Dein, M. Kazerani and M. M. A. Salama, "Optimal photovoltaic array reconfiguration to maximize power production under partial shading," 2012 11th International Conference on Environment and Electrical Engineering (EEEIC), pp. 255-260, 2012.
https://doi.org/10.1109/EEEIC.2012.6221584

[C8] M. Z. Shams El-Dein, M. Kazerani and M. M. A. Salama, "Novel configurations for photovoltaic farms to reduce partial shading losses," 2011 IEEE Power and Energy Society General Meeting, pp. 1-5, 2011.
https://doi.org/10.1109/PES.2011.6039636

[C9] A. A. El-Sattar, N. H. Saad and M. Z. Shams El-Dein, "Modelling and simulation of doubly fed induction generator varable speed wind turbine," 2006 Eleventh International Middle East Power Systems Conference, vol. 2, pp. 492-497, 2006.

Book chapter

[B1] M. Z. Shamseldein and A. Y. Abdelaziz, "Energy Management for Medium-Voltage Direct Current Networks," in Medium Voltage Direct Current Grid, M. M. Eissa, Ed., Academic Press, pp. 43-57, 2019.
https://doi.org/10.1016/B978-0-12-814560-9.00002-1

Public preprints

[P1] M. Shamseldein, "Grid-Mind: An LLM-Orchestrated Multi-Fidelity Agent for Automated Connection Impact Assessment," arXiv:2602.20683, 2026. Preprint.
https://arxiv.org/abs/2602.20683

[P2] M. Shamseldein, "A Hybrid GNN-IZR Framework for Fast and Empirically Robust AC Power Flow Analysis in Radial Distribution Systems," arXiv:2510.04264, 2025. Preprint.
https://arxiv.org/abs/2510.04264

Doctoral thesis

M. Shams El-Dein, Novel Designs for Photovoltaic Arrays to Reduce Partial Shading Losses and to Ease Series Arc Fault Detection, Ph.D. thesis, Department of Electrical and Computer Engineering, University of Waterloo, Canada, 2012.

Patent applications

M. Z. Shamseldein, M. Kazerani and M. M. A. Salama, "Reconfigurable Photovoltaic Structure," U.S. patent application 13/354,876, filed 2012.

M. Z. Shamseldein, M. Kazerani and M. M. A. Salama, "System and Method for Reducing Mismatch in a Photovoltaic Structure," U.S. patent application 13/354,622, filed 2012.

Invited presentation

Invited speaker, IEEE SmartGridComm Workshop, "The Dual Frontier: An Operator's Perspective on AI for Grid Reliability and the AI Demand Challenge," 2025.

Technical methods and tools

AC power flow, contingency assessment, dynamic stability, electromagnetic transients, optimization, uncertainty quantification and machine learning. PSS/E, PSCAD, PowerTech DSA, MATLAB/Simulink, Python, ANDES, pandapower and PyPSA.

Further research foundations

Renewable generation and photovoltaic-array reconfiguration [J14-J19, C4-C9]; medium-voltage DC energy management [B1]; superconducting power delivery [J13]; large-load interconnection governance [J11]. Public preprint work also addresses agent-assisted, multi-fidelity connection assessment [P1]. Current development extends to converter protection and connection assessment, planning and flexibility, and coupled energy systems; these broader directions are not presented as published results.

References

References available upon request.